PAPER_1402_TOO_BIG_TO_FAIL

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PAPER_1402 — UQFF Resolution of the Too-Big-to-Fail Problem (CLOSED — Dispatch Whitepaper)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: D — Cosmology / Galactic Structure (CLOSED)
Date: June 16, 2026
Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)
Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "too_big_to_fail"})`
Calculator surface: `calculate_paradox({"paradox": "too_big_to_fail"})`
Closure helper: `_l96_uqff_axiom_too_big_to_fail_closure()`

The Paradox

ΛCDM predicts the most massive Milky Way subhalos should be too massive to be invisible — yet the most luminous observed dwarfs are not consistent with the densest predicted subhalos. The 'too-big-to-fail' problem.

UQFF Closed Identity

``

```
Subhalo mass-stall ratio = β_i × K_MEX × Φ_res = 0.6029 × (25/12) × 0.84 = 1.055
```

``

$\beta\_i \times K\_MEX \times \Phi\_res = 1.055$ is the F_U_Bi_i buoyancy stall ratio that prevents continued baryonic collapse above the threshold. Subhalos beyond this ratio retain dark mass but suppress observable luminosity.

Physical Interpretation

The most massive subhalos do exist (as predicted) but their baryonic collapse is stalled by SCm buoyancy at the $\beta\_i \times K\_MEX \times \Phi\_res = 1.06$ threshold. They are gravitationally present but optically dim — hence not appearing in the luminous-satellite catalog.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "too_big_to_fail"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_too_big_to_fail_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_too_big_to_fail_closure`
- Dispatch keys: `too_big_to_fail`, `too_big_to_fail_problem`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).
